

No-Go for Observable EFC Background Modifications from BAO–CMB Joint Constraints

Morten Magnusson

Symbiose Research, Sandnes, Norway

ORCID: 0009-0002-4860-5095

DOI: [10.6084/m9.figshare.31985163](https://doi.org/10.6084/m9.figshare.31985163)

April 10, 2026 — Research Note

Abstract. We establish a no-go result for *observationally accessible* background-level modifications in Energy-Flow Cosmology (EFC). Three independent parametrisations — linear $\alpha \cdot T(a)$, exponential $G_{\text{eff}} = G_0 \exp(a_G \cdot S(a))$, and regime-aware piecewise $H(z)$ — are tested against joint Planck 2018 CMB distance priors and DESI DR1 BAO data. All three produce the same structural tradeoff: any $H(z)$ modification at $z < 1$ large enough to improve the BAO fit ($\Delta\chi_{\text{BAO}}^2 \approx -14.6$) is unavoidably detected by CMB distance priors ($\Delta\chi_{\text{CMB}}^2 > +100$), yielding $\Delta\chi_{\text{total}}^2 > +90$ in all cases. The tradeoff is *structural*, not parametric: BAO improvement requires $\delta H/H \sim 2\text{--}3\%$ at $z < 1$, but the CMB angular diameter distance $D_A(z_*)$ integrates through this same redshift interval. We conclude that the EFC background is effectively Λ CDM at current observational precision, and that all entropic modifications large enough to be measured are confined to the perturbation sector (μ, Σ) . This does not exclude the possibility that sub-percent background modifications exist below current sensitivity; it establishes that any modification observable with present data is ruled out. Level 3 (background $H(z)$ recovery) is closed as a null result within current constraints.

1 The Structural Tradeoff

The key observables in background cosmology are all functionals of $H(z)$:

$$D_M(z) = c \int_0^z \frac{dz'}{H(z')}, \quad D_H(z) = \frac{c}{H(z)}, \quad D_A(z_*) = \frac{D_M(z_*)}{1 + z_*} \quad (1)$$

BAO measures D_M/r_d and D_H/r_d at $z \in [0.3, 2.3]$ (DESI DR1, 7 bins). CMB distance priors constrain $\ell_a = \pi D_A(z_*)/r_d$ and $R = \sqrt{\Omega_m H_0^2} D_A(z_*)/c$ at $z_* = 1089.92$.

The critical observation: $D_A(z_*)$ integrates $1/H(z')$ from $z = 0$ to $z = 1089$. Any modification of $H(z)$ at $z < 1$ — regardless of its functional form — contributes to this integral. A fractional change $\delta H/H \sim 2.5\%$ at $z \sim 0.5$ propagates as $\delta D_A/D_A \sim 0.4\%$, which is detectable at $> 5\sigma$ by Planck ($\sigma_{\ell_a} = 0.090$, $\sigma_R = 0.0046$).

2 Three Tests

2.1 Test 1: Linear Coupling $\alpha \cdot T(a)$

The modified Friedmann equation $H^2 = H_{\Lambda\text{CDM}}^2[1 + \alpha T(a)]$, where $T(a) = 1 - \Omega_m(a)$, was tested with Planck MCMC in the CMB Localization paper (DOI: 31368433). Result:

$\alpha = -0.32 \pm 0.40$ (0.82σ), with perfect α - H_0 degeneracy ($r = 0.975$). All joint fits give $|\Delta\chi^2| < 3$. Background gate: **empty**.

2.2 Test 2: Exponential Coupling $G_{\text{eff}} = G_0 \exp(a_G \cdot S)$

From the Track 1 empirical validation, $G_{\text{eff}}(a) = G_0 \exp(a_G \cdot S(a))$ with $a_G = 0.094$ and $S(a)$ the entropy sigmoid ($a_t = 0.55$, $\sigma = 0.10$). The modification is naturally localised to $z < 1$: $S(z = 1.5) < 0.002$, $S(z = 2) < 10^{-4}$.

Table 1: Exponential coupling results (global model, $a_G = 0.094$).

Dataset	$\chi^2_{\Lambda\text{CDM}}$	χ^2_{EFC}	$\Delta\chi^2$
BAO (DESI DR1, 14 pts)	40.80	26.24	-14.56
CMB distance priors (2 pts)	—	—	+112.87
Total	—	—	+98.30

The BAO improvement is significant ($\Delta\chi^2 = -14.6$ for 1 extra parameter). But the CMB penalty is overwhelming. A full (a_G, a_t, σ) grid search over 1875 points finds no combination that achieves $\Delta\chi^2_{\text{BAO}} < -5$ while keeping $\Delta\chi^2_{\text{CMB}} < +5$.

2.3 Test 3: Regime-Aware Piecewise $H(z)$

The EFC regime architecture assigns different physics to different redshift domains: L0 (linear, $z > 2$): $G_{\text{eff}} = G_0$; L1 (structure, $z \sim 0.3$ – 2): G_{eff} modified. A piecewise model was tested:

$$H(z) = \begin{cases} H_{\Lambda\text{CDM}}(z), & z \geq z_{\text{switch}} \\ H_{\text{EFC}}(z), & z < z_{\text{switch}} \end{cases} \quad (2)$$

Result: **identical** to the global model for all $z_{\text{switch}} \geq 1.5$. The sigmoid $S(a)$ is already $< 10^{-3}$ at $z > 1.5$, so the piecewise switch adds no new freedom. The regime transition function *is* the sigmoid.

3 Why Regime Separation Does Not Save the Model

The EFC regime architecture correctly separates *dynamics*: the entropy coupling $S(z = 1089) = 0$ exactly, so CMB-epoch physics is unmodified. However, the CMB *observable* $D_A(z_*)$ is an integral that passes through the L1 regime ($z < 2$) where $H(z)$ is modified.

$$D_A(z_*) = \frac{1}{1+z_*} \int_0^{z_*} \frac{c dz'}{H(z')} = \frac{1}{1+z_*} \left[\underbrace{\int_0^{z_{\text{sw}}} \frac{c dz'}{H_{\text{EFC}}}}_{\text{L1: modified}} + \underbrace{\int_{z_{\text{sw}}}^{z_*} \frac{c dz'}{H_{\Lambda\text{CDM}}}}_{\text{L0: unmodified}} \right] \quad (3)$$

The L1 contribution ($z < 2$) is approximately 15% of the total integral. A 2.5% change in $H(z)$ over this interval produces a $\sim 0.4\%$ shift in $D_A(z_*)$, which exceeds the Planck error budget.

This is an instance of the Regime-Consistent Measurement Principle (RCMP, DOI: 31222900): *regime-separated dynamics does not imply regime-separated observables when the observable integrates across regime boundaries.*

4 The No-Go Statement

No-Go Result. Let $H(z)$ be any modification of ΛCDM in the EFC framework, parametrised by $G_{\text{eff}}(a) = G_0 \cdot f(S(a))$ where $S(a)$ is the entropy sigmoid. If f is chosen to improve the BAO fit by $\Delta\chi_{\text{BAO}}^2 \leq -5$ (requiring $\delta H/H \gtrsim 1.5\%$ at $z \sim 0.5$), then the CMB angular diameter distance shifts by $\delta D_A/D_A \gtrsim 0.2\%$, producing $\Delta\chi_{\text{CMB}}^2 \gtrsim +10$. The joint constraint $\Delta\chi_{\text{total}}^2 = \Delta\chi_{\text{BAO}}^2 + \Delta\chi_{\text{CMB}}^2 > 0$ holds for all tested functional forms of f and all sigmoid parameters (a_t, σ) .

This is not a fine-tuning problem. It is a structural consequence of the fact that BAO and CMB share the low- z portion of the distance integral. The no-go applies to *observable* modifications at current precision; it does not exclude sub-percent background effects that future experiments (e.g., Euclid, DESI DR3) might resolve.

4.1 Scope and Measurement Caveat

This no-go applies specifically to modifications of the background expansion rate $H(z)$. It does not exclude an alternative mechanism: that the BAO preference ($\Delta\chi_{\text{BAO}}^2 = -14.6$) arises from perturbation-level effects that alter how BAO observables are *measured*, rather than how the universe *expands*.

BAO does not measure $H(z)$ directly. It measures galaxy clustering — correlation functions and power spectra — which are then interpreted as distance ratios D_M/r_d and D_H/r_d under the assumption of standard structure formation. If EFC modifies gravitational potentials ($\mu \neq 1$), lensing ($\Sigma \neq 1$), or the growth rate ($f\sigma_8$), these changes can alter the apparent BAO scale without modifying the background. In EFC language: the “speedometer” may read differently even if the “engine” (i.e., $H(z)$) is unchanged.

This possibility — that the BAO signal is a perturbation imprint rather than a background modification — is not tested here and remains open. A full assessment requires forward-modelling BAO observables through an EFC-modified growth history, which is a Level 2 calculation rather than a Level 3 one.

5 Implications for EFC

- (i) **Level 3 is closed at current precision.** The background expansion history $H(z)$ in EFC is indistinguishable from Λ CDM given Planck + DESI DR1 joint constraints. Any modification large enough to be detected is ruled out. This is a result, not a failure.
- (ii) **The ontological status of $S(a)$ is clarified.** At the background level, the entropy field functions as a phenomenological coupling — not a dynamical degree of freedom with its own equation of motion. Whether S has sub-dominant background dynamics below current sensitivity remains an open question that future data may address.
- (iii) **EFC is a perturbation-sector theory with Λ CDM background.** This places it in the same structural class as $f(R)$ gravity in the screening regime, massive gravity with Vainshtein screening, and the EFT of dark energy in the $\alpha_T = 0$ limit. This is a well-established and observationally viable theoretical structure.
- (iv) **All validated EFC results remain intact.** Kill-Test v6, the σ_8 suppression via $\mu(a) < 1$, the KiDS-1000 lensing response, and the SPARC rotation curves are all perturbation-sector results. They do not depend on Level 3 and are unaffected by this null result.
- (v) **The BAO preference ($\Delta\chi^2 = -14.6$) is real but not promotable.** BAO data at $z < 1$ mildly prefer enhanced gravitational coupling, consistent with the perturbation-level $\mu > 1$ signal. This cannot be promoted to a background modification without violating CMB constraints, but it may reflect perturbation-sector effects imprinted on BAO observables — a question for future investigation.

Reproducibility. All code, data, and numerical results are available in the EFC validation repository. The test script (`background_gate_test.py`) uses Planck 2018 distance priors, DESI DR1 BAO data (7 bins), and the Eisenstein–Hu 1998 fitting formula for r_d . The three tests produce consistent results with independent implementations.

References

- [1] Magnusson, M. (2025). Systematic Localization of Late-Time Cosmological Signals in Modified Gravity. Figshare. [10.6084/m9.figshare.31368433](https://figshare.com/10.6084/m9.figshare.31368433)
- [2] Magnusson, M. (2025). EFC Empirical Validation of the Screening Model, Track 1. Figshare. [10.6084/m9.figshare.31940469](https://figshare.com/10.6084/m9.figshare.31940469)
- [3] Magnusson, M. (2026). The Regime-Consistent Measurement Principle (RCMP). Figshare. [10.6084/m9.figshare.31222900](https://figshare.com/10.6084/m9.figshare.31222900)
- [4] Magnusson, M. (2025). A One-Parameter, Four-Channel Consistency Test for EFC Background Coupling. Figshare. [10.6084/m9.figshare.31289806](https://figshare.com/10.6084/m9.figshare.31289806)
- [5] Magnusson, M. (2025). Perturbation-Level σ_8 Suppression via $\mu(a) < 1$. Figshare. [10.6084/m9.figshare.31333414](https://figshare.com/10.6084/m9.figshare.31333414)
- [6] DESI Collaboration (2024). DESI 2024 Results: Baryon Acoustic Oscillations. arXiv:2404.03002.
- [7] Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. *A&A* **641**, A6.